HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Information and Communications Technology

IT4023E – Data Management & Visualization

Paris 2024 Olympic Data Story

A Streamlit dashboard and its visualization techniques

Capstone report: visualization techniques and design rationale

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11,113	206	45	2,268	41
ATHLETES	NOCS	DISCIPLINES	MEDAL RECORDS	VENUES MAPPED

Tool: Streamlit + Plotly Dataset: Paris 2024 Olympic Games

Hanoi, June 2026

Abstract

This report documents an interactive Streamlit data-visualization dashboard built on eight public Paris 2024 Olympic datasets, and analyses how established data-visualization techniques are applied in it, and why each design choice was made. The dashboard turns a flat medal table into a four-act visual story — the global field and where it played out, the medal race, sport specialization, and athlete demographics — using a choropleth, a symbol map, stacked and diverging bars, a heatmap, a treemap, a derived Sankey flow graph, distribution plots, and an animated medal race. For every chart we justify the encoding by data type (nominal, ordinal, quantitative, temporal, spatial) and by well-established criteria — Mackinlay's expressiveness and effectiveness [2], the Cleveland – McGill accuracy ranking [3], Bertin's channels [1], Tufte's integrity rules [7], and Knaflic's storytelling principles [10]. The main contribution is not statistical inference but the deliberate, technique-grounded design of an analytical tool: every figure in this report is generated directly from the same code that produces the live dashboard, so the document and the application are guaranteed to agree.

Team and Contributions

Member	Student ID	Primary responsibilities
Chu Xuan Minh	20235527	Data acquisition, cleaning, and validation.
Nguyen Binh Anh	20235470	Chart design and figure preparation.
Vu Thuy Linh	20235521	Dashboard layout and interaction design.
Nguyen Thanh Vinh	20235576	Report — technique analysis and design rationale.
Nguyen Dinh Truong	20235566	Presentation and overall quality assurance.

Table 1. Division of work by primary responsibility.

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1 Executive Overview

Paris 2024 was the largest, most balanced Summer Olympics on record, and a medal table alone hides most of what made it interesting. This project turns eight public Paris 2024 datasets into an interactive Streamlit dashboard that lets a reader move from the global picture down to a single venue, and this report explains which visualization techniques each chart applies and why.

We deliberately lead with the headline numbers, then narrow into detail — the executive-overview-first arc recommended for data storytelling [13]. The dashboard is organized as four linked story sections; the report mirrors them.

> What the data says, in one paragraph
>
> The United States and China tied at 40 gold medals, with the United States edging the overall table (126 vs. 91 total medals). Host nation France gained the most ground versus Tokyo 2020 (+31 medals) while Japan fell the most (−13). The field reached near gender parity (49.1 % female), the median athlete was 26 years old, and age profiles ranged from a median of 20 in skateboarding and rhythmic gymnastics to 39 in equestrian. Across the Games, 64 athletes were associated with an Olympic or world record.

How to read this report

Section 2 introduces the dataset and the role of visualization. Section 3 documents the data-cleaning work that underpins every chart. Section 4 summarizes the dashboard architecture and story arc. Section 5 is the core: for each major technique it sets out the principle, how that principle appears in a specific chart, and the trade-offs involved. Section 6 collects the analytical findings, and Section 8 covers reproducibility.

2 Dataset and Problem Framing

2.1 Why this dataset

Paris 2024 is the most recent completed Summer Olympics and is easy for any audience to grasp, yet the underlying data are genuinely multi-dimensional: nominal categories (NOC, discipline, venue, gender, medal colour), ordinal ranks (medal-table position, medal hierarchy), quantitative measures (medal counts, delegation size, age, event load), temporal fields (competition start and end dates), and geographic coordinates (venue latitude/longitude, country geometry). This breadth lets us exercise nearly every major visualization family — map, bar, scatter, heatmap, treemap, flow graph, histogram, box plot, and timeline — on a single coherent topic.

The problem visualization solves here is concrete: the public medal table is a one-dimensional ranking that hides geography, sport specialization, athlete demographics, and the spatial footprint of the Games. The dashboard serves all three functions of visualization [4] — it records the Games' results, supports reasoning through inline filters and tooltips, and conveys a story through a guided narrative — and it is information visualization (abstract, structured data), not scientific visualization. The explanatory-versus-exploratory distinction this rests on is a storytelling concept, taken up later in Section 5.

2.2 Data sources and tables

The project draws on two complementary, openly licensed sources, both indexed by and findable on Google Dataset Search:

- "Paris 2024 Olympic Summer Games" (Kaggle) [14]: the official athlete, medallist, athlete–medal, and country medal-table CSVs (the Tokyo comparison comes from the same official tables).

- "Les athletes des Jeux Olympiques de Paris 2024" (data.gouv.fr, Réseau Canopé, ODbL) [15]: an enriched athlete table with age, venue coordinates, competition dates, medal counts, and a record flag — part of the fully Dataset-Search-indexed data.gouv.fr catalogue.

File	Rows	Purpose
`athletes.csv`	11,113	Athlete bios: NOC, gender, birth date, height.
`athletes_disciplines.csv`	11,142	Athlete → discipline membership.
`athletes_events.csv`	14,889	Athlete → event membership.
`medals.csv`	2,268	Athlete–medal records (NOC, athlete, discipline, colour).
`medallists.csv`	2,013	Athlete-level medal totals.
`medal_countries.csv`	92	Paris 2024 country medal table.
`medal_countries_tokyo.csv`	93	Tokyo 2020 comparison medal table.
`paris2024_athletes_enriched.csv`	11,110	Enriched athletes: age, venue coords, dates, records.

Table 2. The eight working CSVs (the first seven built from the official Kaggle dataset, the last from data.gouv.fr). Row counts are verified at load time (header excluded).

3 Data Management and Cleaning

Good visualization rests on disciplined data preparation. Data loading and transformation were implemented as a single automated pipeline and validated against the original source files; the parsing issues below are resolved programmatically rather than by hand-editing the data.

3.1 Parsing quirks resolved

- Semicolon-delimited, French, BOM-encoded enriched table. The data.gouv.fr file is semicolon-delimited, with French headers (Nom, Age, Genre, Medailles d'or, …), a UTF-8 byte-order mark, and 20 columns. It is read accordingly and its columns renamed to a fixed English schema, so the byte-order mark never contaminates a column name.
- Unnamed leading index column. The Tokyo comparison table ships with an unnamed row-number column; it is removed before the Paris–Tokyo comparison.
- French category and value translation. 33 discipline names (Athletisme→Athletics, Natation→Swimming, …) and the gender values (Femme/Homme→Female/Male) are mapped to English so the enriched table aligns with the English-language medal tables. The original French discipline name is retained alongside the English one for traceability.
- Date and type coercion. Birth dates and the day/month/year competition dates are parsed to proper date values; medal counts and coordinates are converted to numbers, with invalid entries set to missing rather than dropped.
- NOC→ISO-3 mapping for the world map. Olympic codes differ from ISO-3 for several countries (GER→DEU, NED→NLD, SUI→CHE); mixed teams (AIN, EOR) have no country geometry and are mapped to "no location" so they are simply omitted from the choropleth.

3.2 Verified shape, types, and completeness

After preparation, all documented counts reconcile exactly: 206 national Olympic committees (of which 92 won at least one medal), 11,113 athletes in the official roster; the demographic and map views draw on a closely matching enriched profile table of 11,110 athletes (age, venue, dates, records) from data.gouv.fr — 2,268 athlete–medal records, and 45 normalized disciplines shared across views. The roster figure headlines the Opening Lens while the athlete-demographic views report the slightly smaller enriched count, and the two are reconciled on screen. Data types are consistent throughout — numeric medal, age, and coordinate fields, proper date fields, and boolean flags for record-holders and for height availability.

The two genuinely incomplete fields are handled honestly rather than imputed:

- Height is missing for 6,033 / 11,110 (54 %) enriched athletes, so height is deliberately not a core metric; a boolean flag records its availability instead.
- Venue coordinates are missing for 1,715 (15 %) rows, almost all "multisite" team-sport entries. Map layers use only rows with valid latitude/longitude (41 venues), but those rows are retained for every non-map chart.

3.3 One conceptual join to watch

Country medal totals and athlete–medal records measure different things and must not be conflated. The country medal table counts official medals by NOC (one gold per event), whereas the athlete–medal table counts each athlete associated with a medal, so a single team gold produces many athlete–medal records. The dashboard keeps these on separate pages: country totals drive the medal table, choropleth, and delta view, while athlete–medal records drive the discipline, heatmap, treemap, and Sankey views. Captions state the unit explicitly so the reader never compares the two scales directly.

4 Dashboard Architecture and Story Arc

The dashboard presents four story sections as inline tabs, and each section places its own controls in the page, beside the charts they drive, so the reader manipulates the data directly. Each chart is generated by a dedicated, stateless chart-building routine fed by a single shared data-preparation layer, so the same cleaned data underpins every view. The four tabs follow a deliberate "broad-to-narrow" arc:

1. Opening Lens — global scale, medal geography, and the Games' venue footprint.
2. Medal Race — who led, and who moved most since Tokyo.
3. Sport Specialization — which NOC dominates which discipline.
4. Athlete Lens — age, gender, workload, medals, and where records fell.

These four sections are the dashboard's story acts. They are presented separately from the technique-by-technique analysis in Section 5.

Following dashboard-design practice (The Big Book of Dashboards [13]), every section opens with prominent KPI cards — large, animated count-up figures that surface the headline numbers — and then answers its question visually in dense multi-column grids. The dashboard deliberately avoids raw data tables: precise values live in chart tooltips (details-on-demand), not in grids dumped onto the page. The Medal Race is crowned by a gold/silver/bronze podium and a medal-efficiency ranking (medals per 100 athletes), and a host-nation spotlight (France) gives the global story a relatable local angle.

5 Visualization Techniques and Design Rationale

This section is the heart of the report. It works through the relevant course chapters in turn; for each it lists the techniques and principles applied, shows how they were applied in a specific chart or page, and records any notes or adjustments.

5.1 Chapter 1: Overview of Data Visualization

Techniques / principles applied

> The role of visualization (1.3) is to amplify cognition (Card, Mackinlay & Shneiderman [4]) through three functions: record information, support reasoning, and convey information to an audience. By type (1.4), this project is information visualization (abstract, structured data), not scientific visualization.

How applied in the dashboard

The Opening Lens opens with a world choropleth (Fig. 1) that conveys the global result before any detail and records every NOC's medal count; hover, zoom, and the inline controls then let it support reasoning — one view serving all three functions.

Notes / adjustments (if any)

We do not use every chart type for its own sake; each family is tied to a specific question, avoiding a decorative "chart gallery". The explanatory/exploratory distinction is taken up later, in the discussion of storytelling.

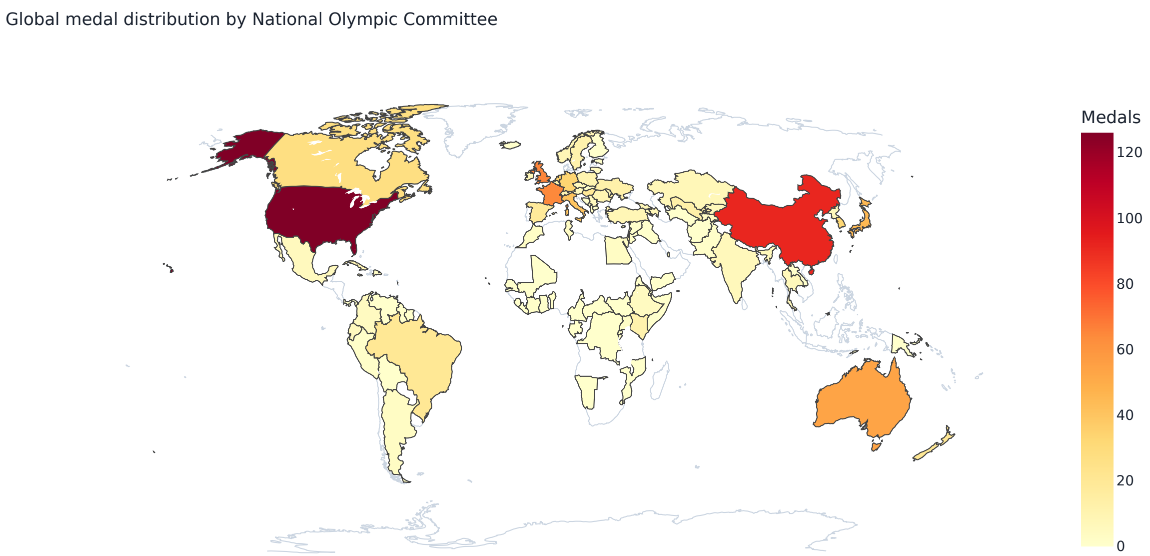

Figure 1. Opening Lens. World choropleth of total Paris 2024 medals by NOC. Colour intensity pre-attentively separates the United States and China from the field. Olympic codes are mapped to ISO-3 so country geometry resolves correctly.

5.2 Chapter 2: Visual Models and Encoding

Techniques / principles applied

> Measurement levels (2.1) — N/O/Q; visual marks (2.2) — points, lines, areas; and visual channels (2.3). Visual encoding (2.4) matches channels to data type by Bertin's levels of organization [1] ($Q \subset O \subset N$) and chosen by Mackinlay's two criteria [2], expressiveness (show all and only the facts) and effectiveness (the most accurately decoded encoding, after the Cleveland–McGill ranking [3]): nominal → hue; ordinal → ordered position or value; quantitative → position / length.

The dataset spans all five standard measurement levels; Table 3 lists them with the channel each is given in the dashboard, following the matching rule above.

Level	Fields in this dataset	How it is encoded
Nominal (N)	NOC, discipline, venue, gender, medal colour	Hue / shape; disambiguated by legend and tooltip
Ordinal (O)	Medal hierarchy (gold > silver > bronze), medal-table rank	Ordered position or colour value (light → dark)
Quantitative (Q)	Medal count, delegation size, age, events entered, medals per 100 athletes	Position / length (primary); area only as a secondary channel
Temporal (T)	Competition start & end dates, Games day	Position on a time axis (daily timeline, competition rhythm)
Spatial (S)	Venue latitude/longitude, country geometry	Map position (symbol map) / region fill (choropleth)

Table 3. The five measurement levels present in the data and the visual channel each receives — nominal to hue, ordinal/quantitative to the perceptually stronger position and length channels.

How applied in the dashboard

The stacked medal bar (Fig. 2, left) is built from area marks and encodes medal count as bar length — the channel Mackinlay's ranking places highest for quantities (effectiveness) — with medal type as a consistent gold/silver/bronze hue. The delegation-vs-medals scatter (Fig. 2, right) uses point marks with horizontal and vertical position for two quantitative variables, area for a third (disciplines entered), and a sequential colour ramp for the ordered medal tier — light to dark, exactly Bertin's rule that value encodes order.

Notes / adjustments (if any)

Carrying four channels in one scatter risks over-encoding; we accept it for a single exploratory relationship view (and keep exact values in the tooltip) rather than splitting it into small multiples. Medal counts and athlete counts are ratio data with a true zero, which is what makes a zero-baseline bar legitimate; the treemap is the Wattenberg "Map of the Market" encoding — area for a quantity, grouping for a category. Medal hues keep a fixed stacking order, so the encoding survives grayscale.

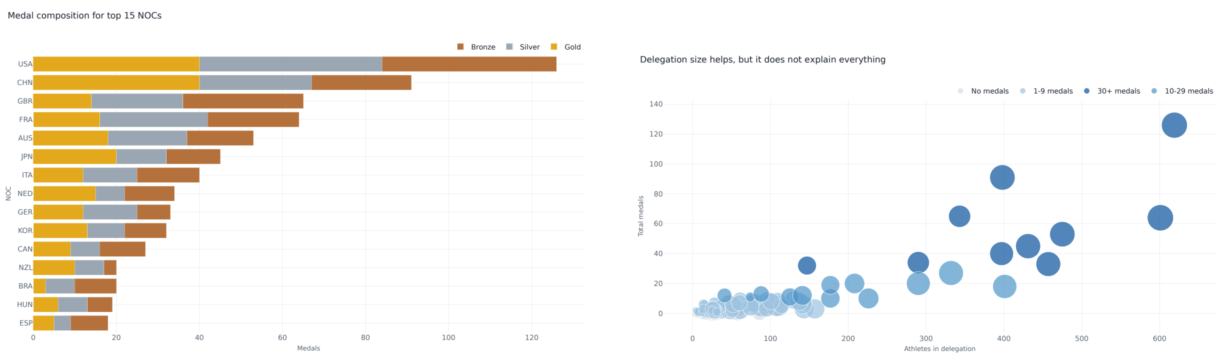

Figure 2. Encoding by data type. Left: medal count as length, medal colour as hue. Right: a bubble scatter where position encodes two quantities, area a third, and an ordered colour ramp the medal tier. Delegation size helps but does not fully explain medal output.

To make the encoding discipline auditable at a glance, Table 4 maps every chart in the dashboard to its variables (with measurement levels), the marks and channels it uses, and the design principle behind the choice. The recurring pattern is deliberate: quantities go to position and length (the most accurately decoded channels), categories to hue, and area is used only where a hierarchy or a third variable justifies it — always with labels, because area is systematically underestimated.

Table 4. Per-chart encoding rationale. Type codes: N nominal, O ordinal, Q quantitative, T temporal, S spatial.

Chart (tab)	Variables (type)	Marks & channels	Encoding rationale
World choropleth (Opening Lens)	NOC (N), total medals (Q), country geometry (S)	Region fill; colour value	Spatial magnitude by region; value encodes order (Bertin); leaders separate pre-attentively.
Medal composition (Medal Race)	NOC (N), count (Q), medal type (O)	Bar length; hue per tier	Length is the most accurate channel for quantity (Cleveland – McGill); fixed hue order survives grayscale.
Athletes vs. medals (Opening Lens)	Athletes (Q), medals (Q), disciplines (Q), tier (O)	x/y position; area; colour value	Position for the two key quantities; area and value carry secondary variables in one exploratory view.
Specialization heatmap (Specialization)	NOC (N), discipline (N), records (Q)	Cell colour value + printed text	Pre-attentive detection of hot cells; redundant text gives exact magnitude (multiple encodings).
Medal-flow Sankey (Specialization)	NOC (N), discipline (N), flow (Q)	Nodes + ribbon width	Bipartite flow graph; ribbon width encodes the weighted edge (records per nation–sport pairing).
Treemap (Specialization)	Discipline (N), country (N), records (Q)	Nested area + colour value	Hierarchy with proportional ink; tile labels offset the lower accuracy of area.
Age distribution (Athlete Lens)	Age (Q), gender (N)	Bar length (freq.); hue	Distribution shape on a zero baseline; gender overlaid for direct comparison.
Gender balance (Athlete Lens)	Discipline (N), gender (N), share (Q)	Bar length (share); hue	100 % stacked: reads part-to-whole share, not raw counts.
Age by discipline (Athlete Lens)	Discipline (N), age (Q)	Box glyph; position (quartiles)	Distribution by group; shows spread without implying sampling error.
Competition load (Athlete Lens)	Events (Q), active days (Q), athletes (Q), medalist rate (Q)	x/y position; area; colour	Aggregated workload cells: bubble area = athlete density, colour = medalist rate; aggregation removes overplotting of the discrete counts.
Medal efficiency (Medal Race)	NOC (N), medals per 100 athletes (Q, ratio)	Bar length	A normalized rate — the honest per-athlete complement to the absolute-count map.
Tokyo delta (Medal Race)	NOC (N), medal change (Q, signed)	Bar length; diverging hue + label	Change around a zero centre; a labelled direction is a second, non-colour cue (CVD-safe).
Venue symbol map (Opening Lens)	Venue (N), lat/long (S), athletes (Q), medal activity (Q)	Map position; size; colour	Point/symbol map; size and colour ride on top of geographic position.
Daily medals (Medal Race)	Games day (T), colour (N), count (Q)	x = time; bar length; hue	Trend over time; the static frame of the animated cumulative race.
Records by discipline (Athlete Lens)	Discipline (N), records (Q)	Bar length	Amounts, ranked — the simplest accurate encoding for a count.

5.3 Chapter 3: Graphical Perception

Techniques / principles applied

> Signal detection (3.1) and Weber's law / JND; magnitude estimation (3.2) — Stevens' power law [6], by which area is underestimated while length is near-unbiased, and the Cleveland & McGill (1984) accuracy ordering [3] (position > length > angle > area), as summarized by Munzner [5]; pre-attentive processing (3.3); multiple encodings (3.4); Gestalt grouping (3.5); and change blindness.

How applied in the dashboard

The country×discipline heatmap (Fig. 3) uses colour intensity so concentrations (USA swimming, Netherlands hockey, GBR rowing) are detected pre-attentively, while printed cell values give exact magnitudes where colour is imprecise. Magnitude comparisons use bars (length), never pie angles — the Cleveland & McGill ordering — and where a quantity is shown by bubble area (scatters, treemap) we cap size and label values, mindful that area is systematically underestimated (Stevens' power law). The Daily / Cumulative toggle and the animated race introduce a frame of motion; pre-attentive colour guides the eye across it to limit change blindness. Gestalt similarity (shared medal hues across pages) and proximity (paired two-column panels) bind related views.

Notes / adjustments (if any)

Cell text is enabled on the heatmap precisely because colour intensity alone is poor for reading exact counts — a direct application of the "multiple encodings" principle.

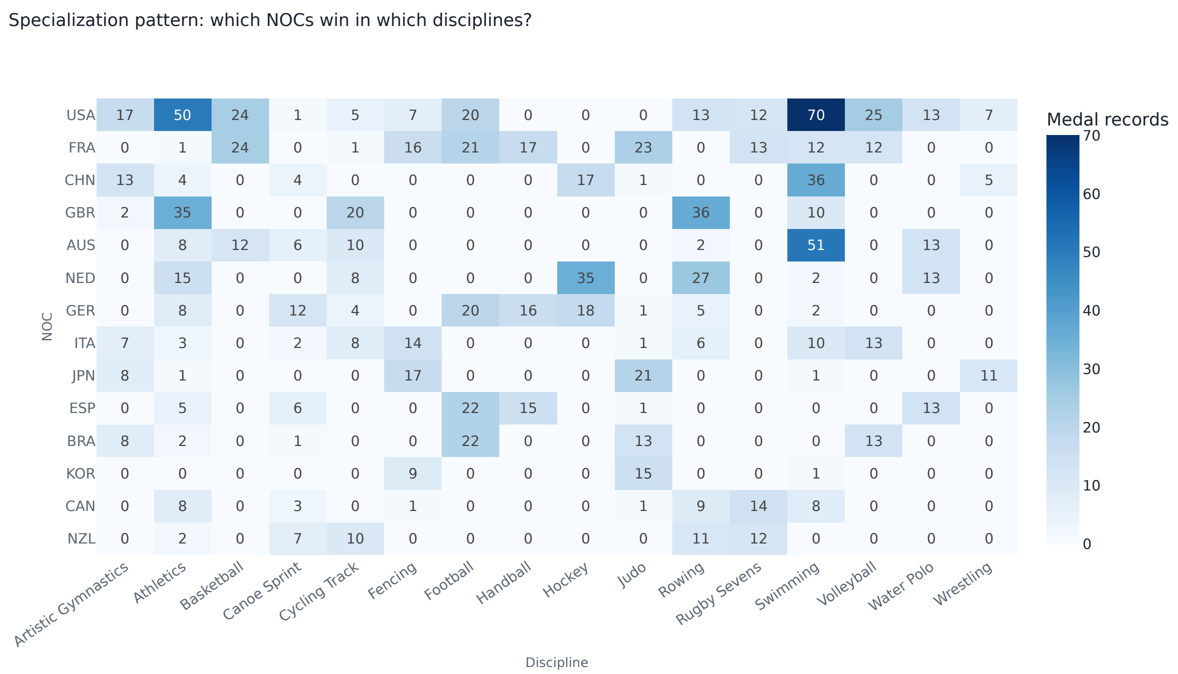

Figure 3. Pre-attentive specialization. Medal-record counts by NOC and discipline. Dark cells (e.g. USA swimming =70) are spotted instantly; the overprinted numbers supply precise magnitudes.

5.4 Chapter 4: Visualization for Multi-dimensional Data

Techniques / principles applied

> The standard chart families map one-to-one onto dashboard pages: amounts (4.2), distributions (4.3), proportions (4.4), relationships (4.5), and trends/change (4.6). Coordinate systems (4.1) are Cartesian throughout; uncertainty (4.7) is addressed below.

How applied in the dashboard

- Amounts: discipline medal-record bars (Fig. 6, left) and the stacked medal-composition bar (Fig. 2, left); the medal ranking is shown as a podium, not a table.
- Distributions: the age histogram by gender (Fig. 4, left) and age box plots by discipline (Fig. 5).
- Proportions: the 100 % stacked gender-balance bar (Fig. 4, right).
- Relationships: delegation size vs. medals (Fig. 2, right), competition load (Fig. 7), and medal efficiency (Fig. 8).
- Trends / change: the Paris-vs-Tokyo delta (Fig. 6, right) and the day-by-day medal timeline (Fig. 12).

Notes / adjustments (if any)

Uncertainty is intentionally not visualized. The data are complete event records, not a sample with sampling error, so error bars would imply a precision that does not exist; in line with best practice [9], we show full distributions (box plots) rather than lossy mean ± error-bar summaries. Trend smoothing (LOESS, moving averages) is likewise omitted: an Olympic calendar of ∼18 discrete daily counts is too short for a meaningful smoother, so the day-by-day timeline shows the raw counts. Coordinate systems are Cartesian throughout — medal totals are long-tailed, so a log axis was considered but rejected to keep the zero-baseline bar reading intuitive for a general audience.

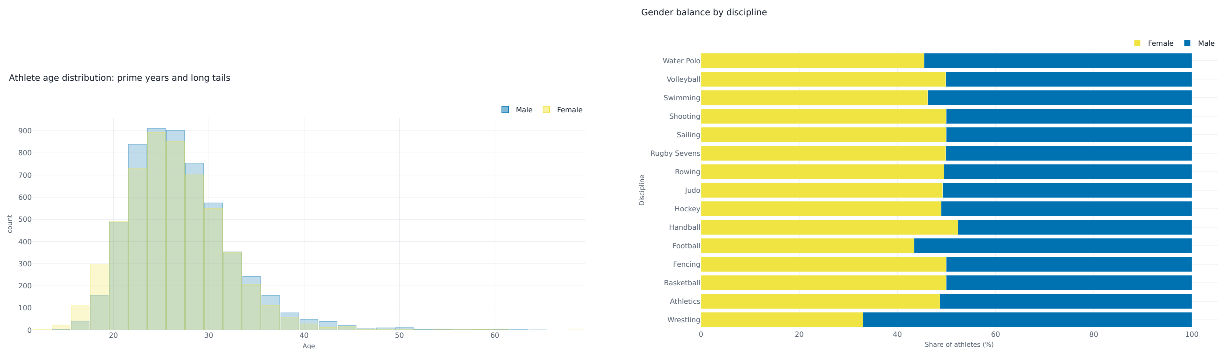

Figure 4. Distributions and proportions. Left: age distribution by gender, overlaid for direct comparison. Right: a 100 % stacked bar reading gender share per discipline, not raw counts.

5.5 Chapter 5: Visualization for Graphs

Techniques / principles applied

> Graph data properties (5.1) — nodes, edges, directionality and edge weight — and graph visualization mechanisms (5.2). The country→discipline relationships form a directed, weighted bipartite graph — rendered as a Sankey flow.

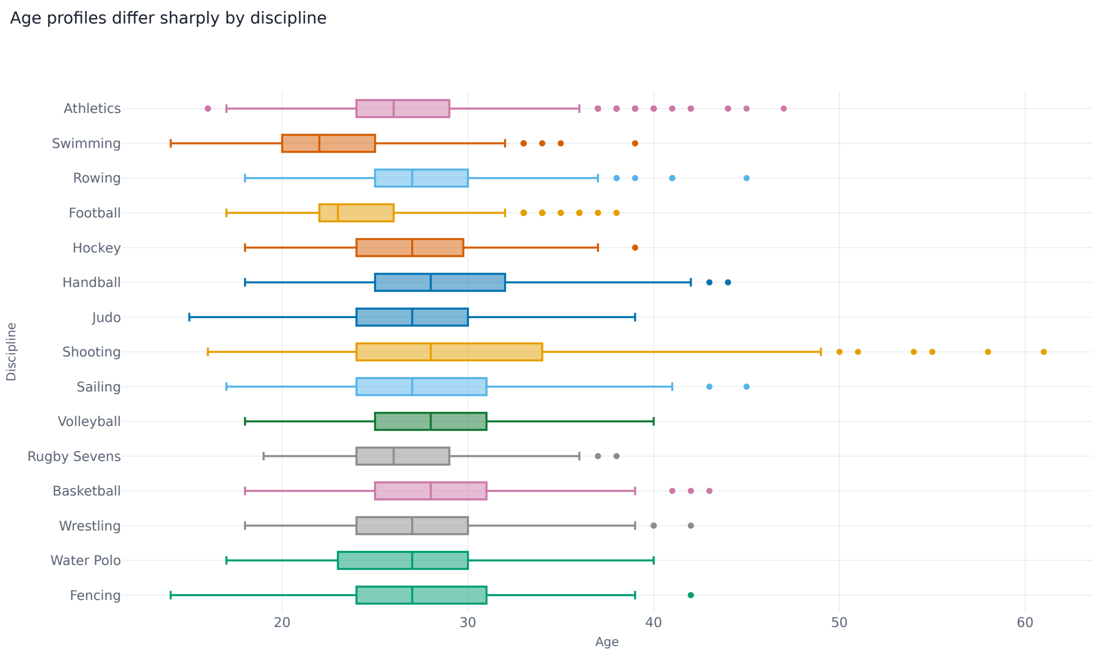

Figure 5. Distribution by group. Age box plots reveal sharply different sport profiles — a median near 20 in skateboarding and rhythmic gymnastics versus 39 in equestrian. Spread is shown without implying sampling uncertainty.

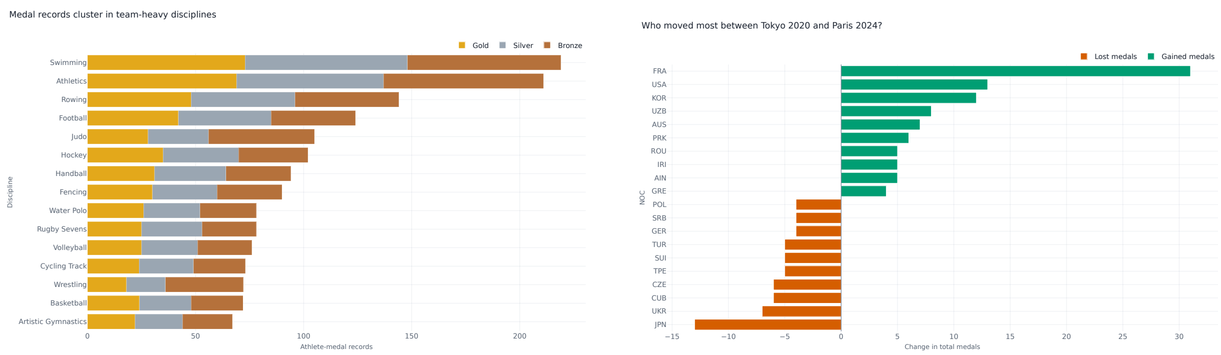

Figure 6. Amounts and change. Left: athlete–medal records per discipline, stacked by colour. Right: a diverging bar of the change in total medals from Tokyo to Paris — France +31, Japan −13.

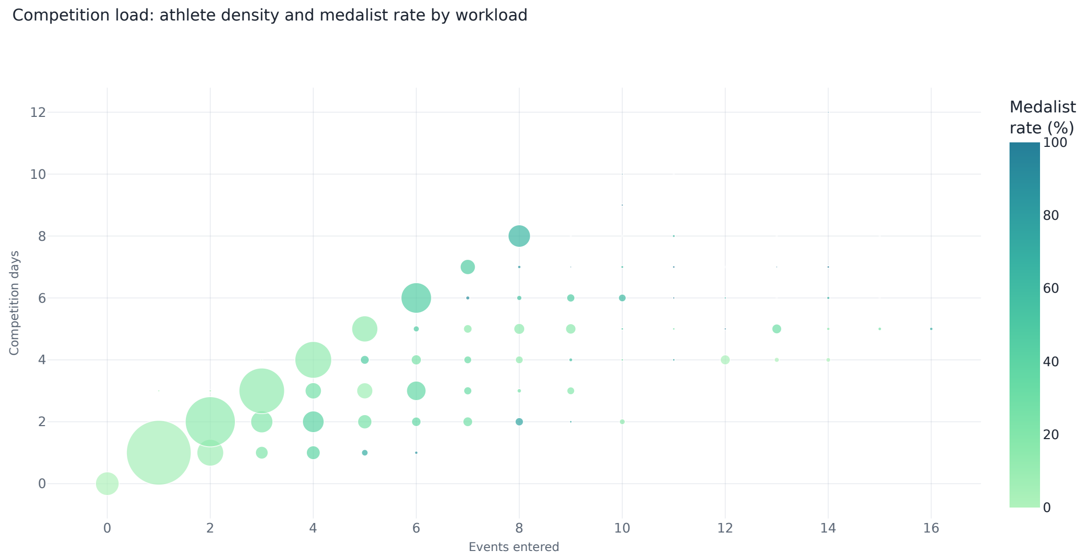

Figure 7. Aggregated workload. Athletes are binned by events entered × active competition days into one bubble per workload cell: bubble area encodes the number of athletes in the cell and colour the cell's medalist rate. Aggregating the discrete counts removes the overplotting an individual scatter would suffer.

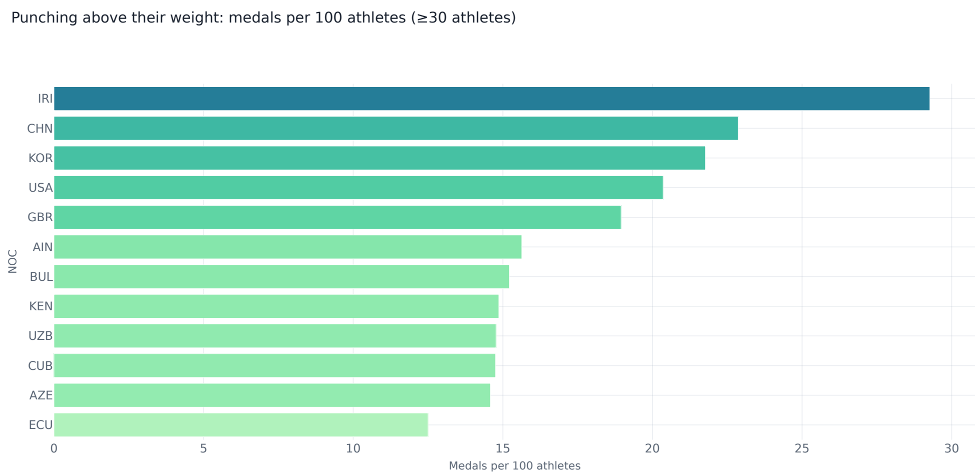

Figure 8. Relationship, normalized. Medals per 100 athletes among delegations of ≥30 athletes — the analytical pay-off of the size-vs-medals scatter. It rewards small, sharp teams that the raw medal table buries, and is the efficiency leaderboard shown in the Medal Race tab.

How applied in the dashboard

The medal-flow Sankey (Fig. 9) treats NOCs and disciplines as two layers of nodes; edge width encodes the number of athlete–medal records flowing from each nation to each discipline. The snap layout keeps the two columns aligned so flow magnitude is read by ribbon thickness, and the widest ribbons mark the strongest nation–sport pairings.

Notes / adjustments (if any)

We are explicit that this is a derived categorical flow graph, not an athlete-to-athlete social network. Node-link layouts contrast with the adjacency matrix [5]; we choose a layered node-link/flow form because this graph is dense by construction (every co-occurring top NOC × discipline) and the question is about flow volume, which ribbon width shows directly. A force-directed layout suits sparse connectivity networks, not aggregated counts, and a matrix would hide the country→discipline path. A third layer to medal colour was considered and dropped: each event awards a near-fixed gold/silver/bronze ratio, so that layer would add a cluttered column carrying no real information. The Sankey thus extends standard flow-map and hierarchy (treemap) mechanisms.

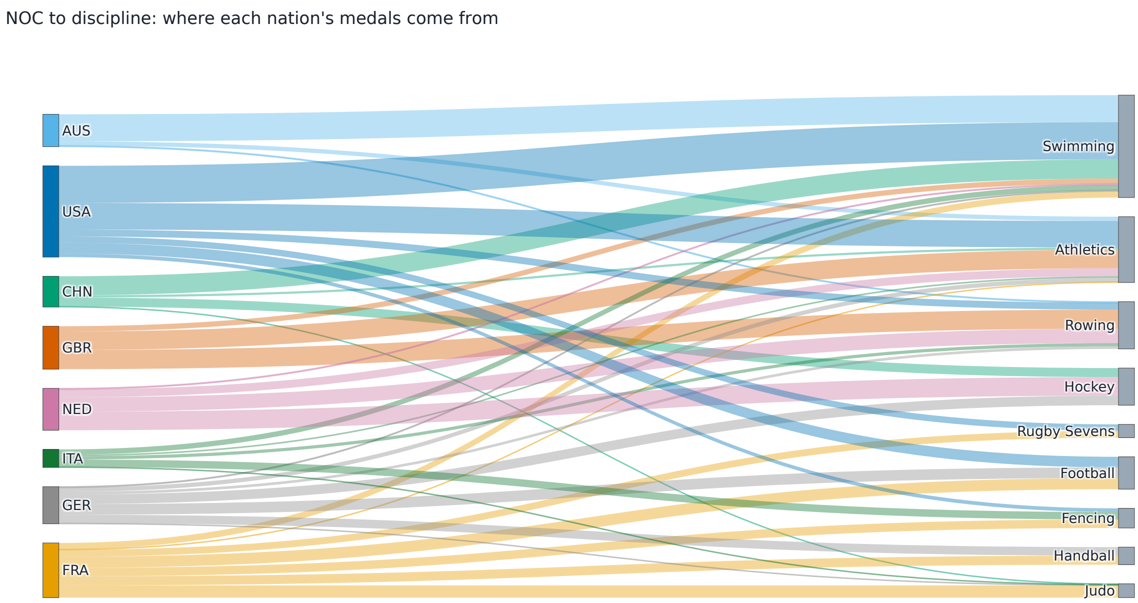

Figure 9. Derived flow graph. NOC → discipline flow. Ribbon width encodes athlete–medal record volume; the two node columns form a bipartite graph, not a connectivity network. The widest ribbons mark the strongest nation–sport pairings.

5.6 Chapter 6: Principles of Figure Design

Techniques / principles applied

> Proportional ink (6.1) [8], handling overlap (6.2), colour use and accessibility (6.3), multi-panel figures (6.4), titles, captions, and tables (6.5), data–context balance (6.6), and avoiding 3D (6.7); plus the integrity rules of graphical honesty — the Lie Factor, maximising the data–ink ratio, and avoiding chartjunk [7]. The chart-choice and figure conventions follow Wilke's Fundamentals of Data Visualization [9].

How applied in the dashboard

- Proportional ink: every bar starts at zero; the treemap (Fig. 10) sizes rectangles strictly by record count.
- Overlap: the workload chart aggregates each discrete (events, days) cell into a single bubble; the heatmap replaces overplotted points with an aggregated grid.
- Colour and accessibility: the qualitative palette is the Okabe–Ito colourblind-safe set [11], reused across pages; the sequential map, heatmap, treemap, and workload scales are perceptually ordered single-hue / Viridis ramps (not Okabe–Ito, which is categorical); medal hues stay constant and luminance-distinct (so they survive grayscale). Where a category is carried by colour alone (medal tier, gender) the palette is CVD-safe and the legend and tooltip disambiguate; gain/loss adds a second, non-colour cue — a labelled direction.
- Graphical integrity (Lie Factor ≈ 1): every bar starts at zero, no axis is truncated, there are no dual axes, and no one-dimensional quantity is encoded as a two-dimensional radius — so the chart cannot overstate an effect. The restrained, KPI-led layout maximises the data–ink ratio and avoids chartjunk (the podium is the one deliberately "useful" embellishment).
- Multi-panel: two-column panels (e.g. Figs. 2, 4) place complementary views side by side.
- Titles & captions: chart titles state a finding ("Delegation size helps, but it does not explain everything"), not just the variables.
- Tables, when used, follow the six rules: good practice is both to "replace a table with a chart" and to set any remaining table well [9], so the dashboard prefers charts (values live on hover, no raw grids), but where this report does tabulate (e.g. Tables 2, 3, 4) it obeys Wilke's six guidelines: no vertical rules, horizontal rules only at the header and boundaries, text columns left-aligned, and number columns right-aligned to equal precision.
- No 3D: every chart is 2D; depth is never used decoratively.

Notes / adjustments (if any)

The treemap is the one area-encoded chart; because area is harder to compare than length, it is used only for hierarchy (discipline→country), with exact values labelled on each tile.

5.7 Chapter 7: Map Visualization

Techniques / principles applied

> Map foundations (7.1) and map types (7.2): a choropleth for region-level intensity and a symbol (bubble) map for point locations, each with its own projection and geocoding concerns.

How applied in the dashboard

The world choropleth (Fig. 1) encodes total medals as fill intensity on a Natural Earth projection, after mapping Olympic NOC codes to standard ISO-3 country codes. The venue bubble map (Fig. 11) places venues by latitude/longitude on a Carto base layer, with bubble size for athlete volume and colour for medal activity — showing how the Games spread from a dense Paris cluster to outlying and overseas venues.

Notes / adjustments (if any)

The choropleth encodes absolute medal totals, which visually over-weights large delegations — a known limitation of choropleths that cartograms address. The honest complement is a per-athlete

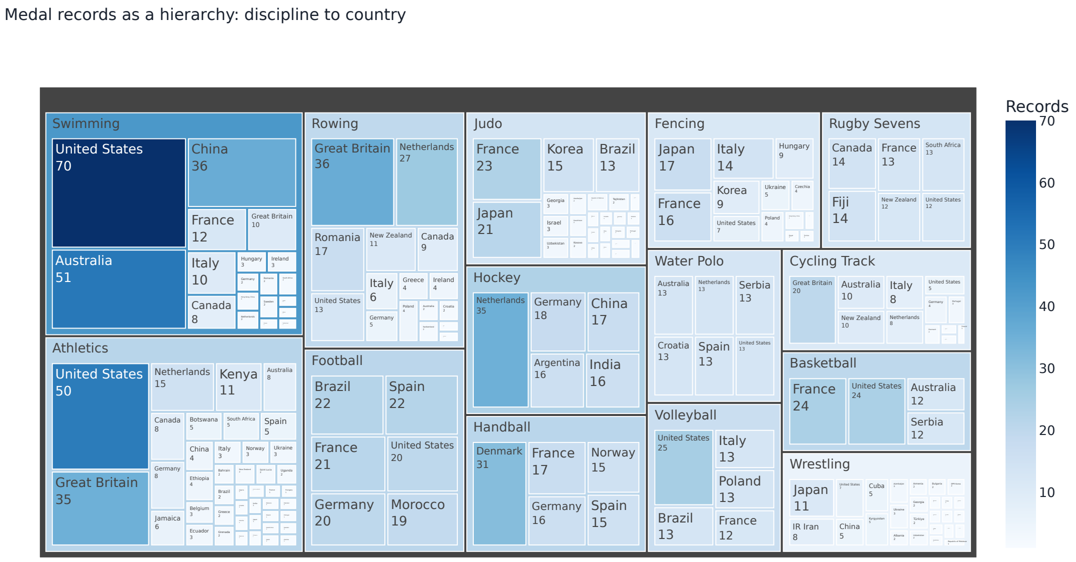

Figure 10. Hierarchy with proportional ink. Medal records nested discipline → country. Tile area is strictly proportional to count and colour reinforces the same magnitude (darker = more records); labels supply exact values to offset the lower accuracy of area judgements.

rate, which we provide as the medals-per-100-athletes efficiency bar (Fig. 8) rather than a second map, since no population data is in scope for a true per-capita choropleth. The venue map's Carto base is Web-Mercator, whose distortion is negligible at France's latitude. Records without coordinates ("multisite" rows) are excluded from map layers but retained everywhere else (§3.3); the choropleth omits mixed teams (AIN, EOR) that have no country geometry.

5.8 Chapter 8: Interactive Visualization

Techniques / principles applied

> Interaction techniques (8.3) — filtering, zoom, selection, and view transformation — together with animation (8.4), details-on-demand, and the tools and libraries used (8.5).

How applied in the dashboard

Interaction is inline and direct. Controls live in the page next to the charts they affect: filtering via segmented-control pills (gender, medal status, medal colours) and multiselects (focus NOCs, disciplines), and view transformation via top-N sliders, an age-range slider, a medal-sort selector that re-ranks charts live, and a Daily / Cumulative toggle on the medal timeline (Fig. 12). The four sections are switched with tabs. Every chart adds details-on-demand via hover tooltips and zoom/pan via the chart toolbar; the maps support drag-pan and scroll-zoom. Each section owns the controls relevant to it, so a reader explores one question at a time. Table 5 maps each interaction to the control that exposes it and the analytical task it serves.

Animation. The Medal Race tab adds an animated bar-chart race: pressing play steps through the Games day by day while each discipline's bar grows with its cumulative medal haul. Animation is used here and only here because the encoded variable — cumulative medals — is genuinely time-ordered, so motion carries information rather than decoration. A static companion view (Fig. 12) shows the same rhythm as daily stacked bars for readers who prefer one frame.

Figure 11. Symbol map. Paris 2024 venues across France. Bubble size encodes athlete volume, colour encodes total athlete medals; the dense Paris cluster contrasts with sailing/surf venues far to the south.

Interaction	Control (where)	Analytical task it serves
Navigate views	Top tabs	Switch between the four story acts.
Filter	Segmented pills	Subset by gender, medal status, or medal colour.
Filter	Multiselects	Compare a chosen set of NOCs or disciplines.
View transform	Top-N slider	Focus on the leaders and cut clutter.
Filter	Age-range slider	Restrict to a demographic sub-population.
Re-order	Medal-sort selector	Re-rank charts live by the chosen metric.
View transform	Daily / Cumulative toggle	Switch the medal timeline's representation.
Animation	Race "play" button	Watch cumulative medals build over the Games.
Details-on-demand	Hover tooltip	Read exact values where a channel is imprecise.
View transform	Zoom / pan	Inspect dense regions of a map or chart.
Filter	Legend toggle	Show or hide an individual series.

Table 5. Interaction-to-task mapping. Every control is inline, beside the chart it drives, so each interaction answers a specific analytical question.

Notes / adjustments (if any)

We deliberately restrict animation to the one time-ordered variable; the rest of the dashboard is a completed-event snapshot where interactive filtering is the more analytical form of dynamics. Every filtered view is verified to degrade to a styled "no data" placeholder rather than crash on an empty selection.

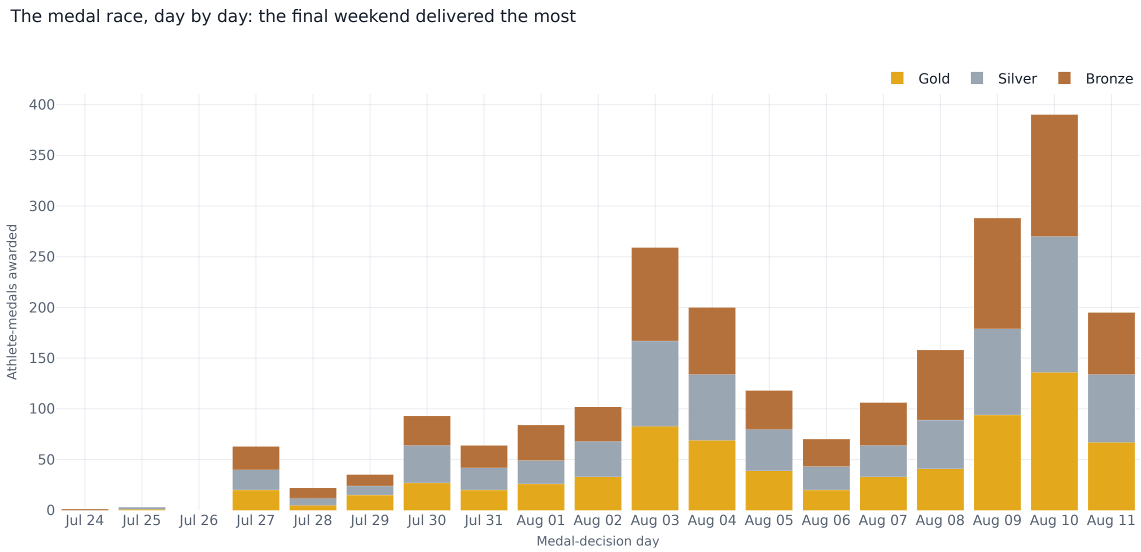

Figure 12. Trend over time. Athlete-medals awarded per Games day, stacked by colour — the static frame of the animated medal race. The tally builds slowly, dips mid-Games, then surges across the closing weekend, peaking on 10 August.

5.9 Chapter 9: Storytelling with Data

Techniques / principles applied

> Fundamental principles (9.2) [10] — the three pillars (data, visualisation, narrative), understand the context (audience and purpose), and the three-act story structure hook → analysis → resolution; storytelling techniques (9.3) — simple-text headline numbers, takeaway titles, redundant encoding; narrative style (9.4) — author-driven vs reader-driven; and interaction and exploration within the story (9.5).

How applied in the dashboard

Understand the context. Before any chart, we fixed the audience and the message. The audience is a general, data-literate classroom, not domain analysts; the single main message is that "the medal table hides most of what made Paris 2024 interesting," and the purpose is to make that case visually and then hand the data over for exploration. Every design choice (headline numbers first, no raw tables, one takeaway per chart) follows from that audience and that message.

Story structure. Each tab is a three-act beat — a hook (a one-line guiding question such as "Who led Paris 2024, and who changed most since Tokyo?"), an analysis that builds the answer in charts, and a resolution in the takeaway title and a spotlight. Across tabs the arc moves macro-to-micro, from the global field to the individual athlete, so abstract totals become concrete and human, ending on the people behind the medals.

Narrative style. The story is author-driven in its fixed tab order and messaged titles, but becomes reader-driven on each page, where filters and tooltips let the reader test their own hypotheses.

The Paris-vs-Tokyo delta is a before/after contrast, and the four linked tabs are progressive disclosure — multiple charts revealing one facet at a time (Bach et al. 2017 [12]) rather than one over-loaded view.

Notes / adjustments (if any)

We apply the standard storytelling devices: simple text (one prominent number with a few supporting words, shown as the KPI cards), takeaway titles that state the finding rather than the variables (the good-vs-bad-title contrast [10]), and redundant encoding (data labels and annotations alongside colour). We also heed the "same data, different stories" caution on storytelling as a double-edged tool: every chart labels its unit (athlete-medals vs official medals) so the framing persuades without misleading. The "executive-overview-first" layout, the podium, and the host-nation (France) spotlight are additionally informed by The Big Book of Dashboards [13]; we replaced every raw data table with a visual, keeping detail on demand in tooltips — a dashboard should show the data, not dump it.

6 Key Insights

1. The gold race was a dead heat; the medal race was not.

The United States and China each won 40 golds, but the United States took 126 total medals to China's 91, driven by depth across many disciplines rather than a few dominant sports (visible in the choropleth and heatmap).

2. Hosting moved the table.

France gained +31 medals versus Tokyo (33→64), the largest swing of any NOC, while Japan fell −13 (58→45). The diverging delta chart makes the host effect legible at a glance.

3. Specialization, not breadth, defines mid-tier powers.

The heatmap shows concentrated strengths — Netherlands in hockey, Great Britain in rowing, Australia in swimming — whereas the top two NOCs spread medals broadly.

4. The field reached near gender parity, with sport-specific age profiles.

Women made up 49.1 % of athletes; the median athlete was 26, but discipline medians ranged from 20 (skateboarding, rhythmic gymnastics) to 39 (equestrian) — a spread the box plots surface immediately.

5. The Games were geographically dispersed.

Although venues cluster in Paris, the symbol map shows competition reaching far south (sailing in Marseille, plus overseas surfing), reinforcing that Paris 2024 was a spatial event, not a single-city one.

7 Conclusion and Future Work

The project set out to show that a flat medal table hides most of what made Paris 2024 interesting, and the dashboard makes that case visually: the global choropleth, the Tokyo delta, the specialization heatmap, the athlete distributions, and the venue map each surface a facet — depth, host effect, specialization, demographics, dispersion — that a ranked table cannot. Just as importantly, the build is a deliberate exercise in technique: every chart's encoding is chosen by data type and justified against established criteria (Table 4), colour is the Okabe–Ito colourblind-safe set used only to carry meaning, the figures keep a Lie Factor of about one, and the four-act arc applies recognized storytelling principles [10].

We are deliberate about the limits. The dashboard reports complete event records, not a sample, so it shows full distributions rather than error bars; the choropleth encodes absolute medal totals, with the medals-per-100-athletes bar standing in for the per-capita rate a cartogram would otherwise provide; and athlete height is too sparse (54 % missing) to be a core metric. Natural next steps follow directly: bring in national-population data for a true per-capita choropleth or cartogram; add a live data-refresh path so the same pipeline serves future Games; run a short colour-blindness and usability check with real users; and, where future data carries genuine sampling, add the uncertainty layer we currently and correctly omit.

8 Reproducibility

Every figure in this report is generated directly from the same code and the same prepared data that produce the live dashboard — there are no screenshots and no separately drawn figures — and the document is typeset from those vector outputs. Each figure shown here is therefore identical to what a reader sees in the application, and the report cannot drift out of step with the dashboard.

The dashboard was exercised systematically across all four sections and every interactive control, including empty and narrowly filtered selections; each of these degrades gracefully to a styled "no data" state rather than failing. The row counts, data types, and missing-value rates reported in §3 were checked against the underlying data and reconcile exactly. Together these checks make the figures trustworthy: they are reproducible from the source data and consistent with the application that produced them.

Data: the official Paris 2024 CSVs (Kaggle "Paris 2024 Olympic Summer Games") and the data.gouv.fr enriched athlete table (ODbL), both findable on Google Dataset Search. Built with Streamlit and Plotly; figures are generated directly from the dashboard's own charting code.

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